

FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

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PROFESSIONAL PROFILE

Computer Graphics Engineer specialized in real-time rendering, procedural generation, and tool development. Graduated with honors in Cinema and Media Engineering, focusing on XR and graphics programming. Experienced in writing custom shaders, optimizing GPU performance, and building pipelines that bridge engineering and art.

EDUCATION

M.Sc. Cinema and Media Engineering - Politecnico di Torino **2024 - 2026**

- **Expected Grade:** 110/110 Cum Laude (Graduating Dec 2026)

B.Sc. Cinema and Media Engineering - Politecnico di Torino **2021 - 2024**

- **Final Grade:** 100/110

WORK EXPERIENCE

Lemons In The Room | Florence, Italy **03/2026 - Present**

XR Master Thesis - R&D Graphics Programmer

- Built a real-time HLSL Raymarching framework for Meta Quest 3 to render Signed Distance Fields (SDFs).
- Integrated BLE data to drive procedural geometry changes based on physiological signals.
- Implemented Dynamic Bounding Volumes and Foveated Raymarching, ensuring a stable 72Hz framerate on mobile VR hardware.

Level Up Lab | Turin, Italy **01/2025 - Present**

Technical Artist & Gameplay Programmer

- Programmed core gameplay mechanics and physics interactions in C# (Unity) for the "Line Operator" project.
- Built the 3D asset pipeline, integrating models and animations directly into the engine.
- Wrote custom HLSL shaders and post-processing volumes to match the art direction.

Politecnico di Torino | Turin, Italy **10/2024 - 01/2025**

Teaching Assistant (GUI)

- Tutored students in UI/UX principles and frontend programming during weekly laboratory

sessions.

- Wrote technical documentation and code examples used by the class.

RKH SRL | Turin, Italy

09/2023 - 06/2024

Video Artist

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted with lighting setups on set.
- Edited footage and managed post-production workflows using DaVinci Resolve.

SKILLS

- **Languages:** C#, C++, Python, HLSL/GLSL
- **Engines:** Unity, Godot, Unreal Engine 5
- **Software:** Blender, DaVinci Resolve, Jira, Git
- **Methodologies:** Agile, Tech Direction, Procedural Workflows

PERSONAL PROJECTS

- **Harmony VR (Avenue Cove):** Built an audio-reactive VR experience in Unity. Linked real-time audio frequencies to VFX Graph and custom shaders to drive particle simulations.
- **UE5 Tech Art Pipeline:** Developed procedural environments using PCG and Auto-Materials. Scripted a homing projectile system in Niagara using Point Attraction Forces. Integrated MetaHumans with custom groom collision physics.